

Hypertension Risk Assessment Using Machine Learning on Philippine Nutrition Survey Data

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Abstract

This project develops a web-based hypertension risk assessment tool using the 2015 Philippine National Nutrition Survey (NNS) datasets. The system integrates clinical, dietary, and anthropometric features to estimate an individual's probability of hypertension. The repository contains both:

1. A Streamlit deployment stack for end-user risk assessment.
2. A reproducible unified training pipeline that provides a consolidated execution path with notebook-faithful defaults.

Important: the unified pipeline does not replace or invalidate the original notebooks. `EXP 0` and `EXP A` through `EXP F` remain valid, runnable, and official experiment artifacts.

Repository Structure

```
attestation/
container/
journal article/
manual/
Proposal Docs/
SP Docs/
src/
  Dockerfile
  requirements.txt
  thesis_webapp/
    app.py
    backend.py
    app_constants.py
    counterfactuals.py
    explainability.py
    styles.py
  training/
    EXP 0 Create Merge Dataset.ipynb
    EXP A KNN_RF.ipynb
    EXP B XGB_AdaBoost.ipynb
    EXP C LogReg_CatBoost_LightGBM.ipynb
    EXP D_Naive Bayes.ipynb
    EXP E EDA.ipynb
    EXP F COMPILE_RESULTS.ipynb
    EXP unified_pipeline.py
README.md
```

System Overview

The Streamlit web app accepts patient data across four domains:

Domain	Examples
Demographic / Clinical	Age, sex, ethnicity, blood pressure history

Behavioral	Smoking status, alcohol consumption, binge drinking
Anthropometric	Weight, height, waist circumference, hip circumference
Dietary	Intake across 27 food groups (grams), derived energy and macronutrient totals

The app then:

1. Preprocesses inputs through the saved preprocessing artifacts.
2. Produces calibrated hypertension probabilities.
3. Applies the selected operating threshold from training outputs.
4. Provides SHAP and LIME explainability outputs.
5. Generates counterfactual recommendations.

Machine Learning Experiments

Experiment	Models
EXP A	K-Nearest Neighbors, Random Forest
EXP B	XGBoost, AdaBoost
EXP C	Logistic Regression, CatBoost, LightGBM
EXP D	Naive Bayes

All experiments use a two-stage search plus post-calibration evaluation flow.

Unified Training Pipeline (Notebook Parity)

`EXP unified_pipeline.py` is the consolidated Python execution path for repeatable training runs.

It is intended to mirror notebook behavior while keeping the notebook workflow intact. `EXP 0` and `EXP A` through `EXP F` can still be run independently and are not deprecated.

Main script:

```
python "src/training/EXP unified_pipeline.py" --root src/training --out-dir exp3_unified
```

Default behavior is strict notebook-faithful execution:

1. Per-model Stage 1 and Stage 2 budgets come from the original EXP A-D configurations.
2. Threshold grid defaults to `np.round(np.arange(0.35, 0.70, 0.05), 2)`.
3. Search spaces match notebook search spaces for KNN, RF, XGBoost, AdaBoost, LogReg, CatBoost, LightGBM, and Naive Bayes.
4. Global stage CLI flags are optional overrides (`--s1-trials`, `--s1-folds`, `--s1-epochs`, `--s2-refine`, `--s2-folds`, `--s2-epochs`, `--top-k-s1`, `--top-k-s2`, `--final-epochs`). If omitted, notebook parity defaults are used.

Example override run:

```
python "src/training/EXP unified_pipeline.py" --root src/training --out-dir exp3_override --s1-trials 20 --s2-refine 3
```

Primary outputs (under `src/training/<out-dir>/`):

- `stage1_summary.csv`
- `stage2_summary.csv`
- `pre_calibration_results.csv`
- `post_calibration_results.csv`
- `models/best_model_bundle.joblib`
- `plots/` and explainability artifacts

Dataset

- Source: 2015 Philippine National Nutrition Survey (NNS), Food and Nutrition Research Institute - DOST (FNRI-DOST)
 - Domains: Clinical, Dietary, Anthropometric
 - Features: Demographic, clinical, diet, body measurements, and behavior variables
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Technologies

Category	Libraries / Tools
Web framework	Streamlit
ML / preprocessing	scikit-learn, XGBoost, CatBoost, LightGBM, imbalanced-learn
Calibration	venn-abers
Explainability	SHAP, LIME
Data	pandas, NumPy, SciPy
Container	Docker (Python 3.11-slim), port 8501

Running the Application

See [manual/DEVELOPMENT.md](#) for full instructions.

Quick start (Docker):

```
cd src
docker build -t hypertension-app .
docker run -p 8501:8501 hypertension-app
```

Open: `http://localhost:8501`

Usage

See [manual/HELP.md](#) for the user manual.

Attestation

The signed ORI Attestation Form is in `attestation/`.

License

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